HeartCycle: X-point signed reference errors · Overall
n=17 participants; 371 strict common-reference beats · complete range; linear centre ±150 ms

Negative (early) · Positive (late) · Zero error

X1 · First local minimum

X2 · Global minimum

X3 · d3 upward zero

X4 · d3 minimum

X5 · 45% downstroke

X6 · 25% downstroke

X7 · d2 upward zero

X8 · d2 after minimum

X9 · d2 minimum

X10 · Weissler HR

X11 · Adaptive half descent

X12 · RR-adaptive minimum

X13 · 15% downstroke

Heartbeats (count)

Earlier ← signed detected − reference error (ms) → Later